Applied Statistics (MATH10096)

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Prediction

- Having fitted a model, often we would like to use the model to make predictions for a set of predictor variables values (which may have been used for fitting, or may be new).
- The procedure to obtain such predictions/predicted values is as follows:
  1. Use the new predictor variables values to create a vector $\mathbf{x}_0$ in exactly the same way as the predictors would have been used to create a row of the design matrix if they had been used for model fitting.
  2. The prediction is then $\hat{\mu}_0 = \mathbf{x}_0^{\mathsf{T}}\hat{\boldsymbol{\beta}}$.
  3. Note that $\hat{\mu}_0$ is an estimator of the mean $\mathbb{E}(y_0) = \mathbf{x}_0^T\boldsymbol{\beta} = \mu_0$ of the new observation y_0 with predictors $\mathbf{x}_0$.

Prediction

- For example, the finally selected tyre wear model (by backward selection through F-testing) was:

$$y_i = \beta_1 + \beta_2 t_i + \beta_3 t_i h_i + \beta_4 t_i^2 + \beta_5 h_i^2 t_i + \beta_6 t_i^3 + \beta_7 h_i^3 + \epsilon_i.$$

- Hence to make a prediction for (h_0, t_0), we form the vector $\mathbf{x}_0^{\mathrm{T}} = (1, t_0, t_0 h_0, t_0^2, h_0^2 t_0, t_0^3, h_0^3)$, and the prediction is $\hat{\mu}_0 = \mathbf{x}_0^{\mathrm{T}} \hat{\boldsymbol{\beta}}$.
- With the values, say, $h_0 = 50$ and $t_0 = 200$ then $\mathbf{x}_0^{\mathrm{T}} = (1, 200, 10000, 40000, 500000, 8000000, 125000)$.

```
x0 <- c(1, 200, 10000, 40000, 500000, 8000000, 125000)
t(x0)%*%coef(m4)
```

```
##           [,1]
## [1,] 236.0129
```

Prediction

↪ The `R` function `predict` automates the calculation of predictions.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50))
##         1
## 236.0129
```

↪ This predicted value represents the estimated mean abrasion loss of a tyre with an hardness of 50 and a tensile strength of 200.

Prediction

- Of course, we need more than just a point estimate to make informed choices or decisions and this brings us to the task of deriving a confidence interval for μ_0.
- First, note that
$$\mathbb{E}(\hat{\mu}_0) = \mathbb{E}(\mathbf{x}_0^T \hat{\boldsymbol{\beta}}) = \mathbf{x}_0^T \mathbb{E}(\hat{\boldsymbol{\beta}}) = \mathbf{x}_0^T \boldsymbol{\beta} = \mu_0.$$
- The variance of $\hat{\mu}_0$ is also easy to obtain
$$\begin{aligned} \text{var}(\hat{\mu}_0) = \text{var}(\mathbf{x}_0^T \hat{\boldsymbol{\beta}}) &= \mathbf{x}_0^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0 \\ &= \mathbf{x}_0^T \mathbf{V}_{\hat{\boldsymbol{\beta}}} \mathbf{x}_0, \qquad \mathbf{V}_{\hat{\boldsymbol{\beta}}} = \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2 = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\sigma^2. \end{aligned}$$
- Because $\hat{\boldsymbol{\beta}}$ is normally distributed, we have that $\hat{\mu}_0 = \mathbf{x}_0^{\mathrm{T}} \hat{\boldsymbol{\beta}}$ is also normally distributed,
$$\hat{\mu}_0 \sim N(\mu_0, \sigma^2_{\hat{\mu}_0}), \quad \text{with} \quad \mu_0 = \mathbf{x}_0^{\mathrm{T}} \boldsymbol{\beta}, \quad \sigma^2_{\hat{\mu}_0} = \mathbf{x}_0^{\mathrm{T}} \mathbf{V}_{\hat{\boldsymbol{\beta}}} \mathbf{x}_0.$$
- Standardising and replacing σ^2 by $\hat{\sigma}^2$ yields
$$\frac{\hat{\mu}_0 - \mu_0}{\hat{\sigma}_{\hat{\mu}_0}} \sim t_{n-p}.$$

Prediction

↪ So an $100(1-\alpha)\%$ CI for μ_0 is

$$\hat{\mu}_0 \pm t_{1-\alpha/2,n-p}\hat{\sigma}_{\hat{\mu}_0}.$$

↪ The standard error of the prediction as well as the confidence interval are easily obtained in R.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), se = TRUE)
```

```
## $fit
##        1
## 236.0129
##
## $se.fit
## [1] 16.64521
##
## $df
## [1] 23
##
## $residual.scale
## [1] 27.67362
```

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), interval = "confidence")
```

```
##        fit      lwr      upr
## 1 236.0129 201.5797 270.4461
```

Prediction

- The confidence intervals for predictions, considered so far, are random intervals that have some specified probability of including the true expected response corresponding to some predictor variable values.

- Sometimes it is useful to obtain prediction intervals which are intervals within which we would expect a new independent observation of a response variable, say y_0, to lie, with some specified probability, given some particular values for the predictors, say $\mathbf{x}_0$.

- In the context of the tyre wear example, this means we would be interested in knowing the range of reasonable abrasion loss values for a tyre with some given characteristics (i.e., for a specific hardness and tensile strength, e.g., 50 and 200, respectively).

Prediction

- To construct a prediction interval for a new observation y_0, we consider the residual of that observation, also known as the prediction error, $\hat{\epsilon}_0 = y_0 - \mathbf{x}_0^{\mathsf{T}}\hat{\boldsymbol{\beta}} = y_0 - \hat{\boldsymbol{\mu}}_0$, where

$$y_0 = \mu_0 + \epsilon_0 = \mathbf{x}_0^{\mathsf{T}}\boldsymbol{\beta} + \epsilon_0, \quad \epsilon_0 \sim N(0, \sigma^2).$$

- We have that

$$\mathbb{E}(\hat{\epsilon}_0) = \mathbb{E}(y_0 - \hat{\mu}_0) = \mathbb{E}(y_0) - \mathbb{E}(\hat{\mu}_0) = \mu_0 - \mu_0 = 0,$$
$$\text{var}(\hat{\epsilon}_0) = \text{var}(y_0) + \text{var}(\hat{\mu}_0) = \sigma^2 + \sigma^2_{\hat{\mu}_0}.$$

- When calculating the above variance, we have used the following facts:

1. The independence of y_0 and $\hat{\mu}_0$ (since y_0 is independent of the observations $y_1, \ldots, y_n$ used to estimate $\hat{\boldsymbol{\beta}}$, and thus $\hat{\mu}_0 = \mathbf{x}_0^{\mathsf{T}}\hat{\boldsymbol{\beta}}$).
2. y_0 follows the same data-generating mechanism as the other observations, and therefore its variance is σ^2.

Prediction

- Furthermore,
$$\hat{\epsilon}_0 \sim N(0, \sigma^2 + \sigma^2_{\hat{\mu}_0}).$$
- Standardising and replacing $\hat{\sigma}^2$ for σ^2 yields
$$\frac{y_0 - \hat{\mu}_0}{(\hat{\sigma}^2 + \hat{\sigma}^2_{\hat{\mu}_0})^{1/2}} \sim t_{n-p},$$
- Therefore
$$\Pr\left(\hat{\mu}_0 - t_{1-\alpha/2,n-p}(\hat{\sigma}^2 + \hat{\sigma}^2_{\hat{\mu}_0})^{1/2} < y_0 < \hat{\mu}_0 + t_{1-\alpha/2,n-p}(\hat{\sigma}^2 + \hat{\sigma}^2_{\hat{\mu}_0})^{1/2}\right) = 1 - \alpha$$
- That is, a $100(1-\alpha)\%$ prediction interval for a new response observation at a given set of predictor values $\mathbf{x}_0$ is

$$\hat{\mu}_0 \pm t_{1-\alpha/2,n-p}(\hat{\sigma}^2 + \hat{\sigma}^2_{\hat{\mu}_0})^{1/2}.$$

- Per construction, the prediction interval is always wider than the corresponding confidence interval for μ_0.

Prediction

- Predictions intervals are easily obtained in R.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), interval = "prediction")
##        fit     lwr      upr
## 1 236.0129 169.208 302.8178
```

- Notice that this is much wider than the corresponding confidence interval for the predicted mean response, which was $(201.5797, 270.4461)$.

- And it has to be, as this is the interval within which 95% of new replicate observations should lie (while the confidence interval was concerned with bracketing the long term average of such replicate observations).